Auscultate Lungs

No lung disease (normal lungs)

- Clear lung sounds
- No Chest Trauma
- Equal Air Entry

Setting	Normal Lungs
MODE	IPPV/SIMV
Frequency (Respiratory Rate)	12 – 15
Tidal Volume	6-8mL/kg
Pmax	25 – 30
PEEP	5
I:E	1:2
Fi02	1.0

Obstructive Lung Disease

- Wheezing
- Asthma/COPD
- No breath sounds/silent chest
- Anaphylaxis

Setting	Obstructive Lungs
MODE	IPPV/SIMV
Frequency (Respiratory Rate)	8 – 10
Tidal Volume	6-8mL/kg
Pmax	35 – 50
PEEP	3-5
I:E	1:3 – 1:4
Fi02	1.0

Restrictive Lung Disease Likely

- Crackles
- Pulmonary Oedema
- Pulmonary contusion (e.g. chest trauma)
- ARDS (e.g. sepsis)
- Pneumonia

Setting	Restrictive Lungs
MODE	IPPV/SIMV
Frequency (Respiratory Rate)	15 – 25
Tidal Volume	4-6mL/kg
Pmax	30 – 40
PEEP	5 – 15
I:E	1:2
Fi02	1.0

Cardiac Arrest (During CPR)

Setting	CPR
MODE	IPPV / Manual
Frequency (Respiratory Rate)	10
Tidal Volume	6 – 8mL/kg
Pmax	30 – 35
PEEP	3
I:E	1:2
Fi02	1.0

Patient Care Goals

- Maintain adequate oxygenation.
- Maintain adequate minute ventilation and capnography targets based on patient pathophysiology.
- Prevent or limit risk of short- and long-term invasive airway and ventilator-associated complications including barotrauma, pneumothorax, aspiration, over-ventilation.
- Provide advanced airway support for patients with obstructive, restrictive, or no lung disease, tailoring ventilation settings to the underlying pathology.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients requiring invasive ventilation due to cardiac arrest (during CPR), obstructive lung disease (e.g., asthma, COPD, anaphylaxis), restrictive lung disease (e.g., pulmonary edema, ARDS, sepsis, pneumonia), or no lung disease (normal lungs).

Exclusion Criteria

- Patients not requiring a ventilator.
- Patients requiring the ventilator for non-invasive ventilation (CPAP/BiPAP).

Key Patient Assessment Points

1. Auscultate lungs to determine underlying lung pathology:
    - No lung disease (clear lung sounds, no chest trauma, equal air entry).
    - Obstructive lung disease (wheezing, asthma/COPD, no breath sounds/silent chest, anaphylaxis).
    - Restrictive lung disease (crackles, pulmonary edema, chest trauma, ARDS, sepsis, pneumonia).
2. Initiate continuous assessment and monitoring with: SpO2, ECG, and EtCO2 to assess oxygenation, cardiac rhythm, and ventilation status.

Patient Management and Interventions

1. **Ventilator Settings Preparation**
    - Select initial settings and adjust to meet the patient's needs (all settings available, e.g., I:E ratio). If a setting is not available, consider utilizing a Bag Valve Mask instead.
    - **Normal Lungs:**
        - MODE: IPPV/SIMV
        - Frequency (Respiratory Rate): 12-15
        - Tidal Volume: 6-8 mL/kg IBW
        - Pmax: 25-30
        - PEEP: 5
        - I:E: 1:2
        - FiO2: 1.0
    - **Obstructive Lungs:**
        - MODE: IPPV/SIMV
        - Frequency (Respiratory Rate): 8-10
        - Tidal Volume: 6-8 mL/kg IBW
        - Pmax: 35-50
        - PEEP: 5
        - I:E: 1:3-1:4
        - FiO2: 1.0
    - **Restrictive Lungs:**
        - MODE: IPPV/SIMV
        - Frequency (Respiratory Rate): 15-25
        - Tidal Volume: 4-6 mL/kg IBW
        - Pmax: 30-40
        - PEEP: 5-15

    - I:E: 1:2
    - FiO2: 1.0
  - **CPR (Cardiac Arrest):**
    - MODE: IPPV/Manual
    - Frequency (Respiratory Rate): 10
    - Tidal Volume: 6-8 mL/kg IBW
    - Pmax: 30-35
    - PEEP: 5
    - I:E: 1:2
    - FiO2: 1.0

2. **Invasive Ventilation Procedure**
   - After the patient has been stabilized on the ventilator, adjust FiO2 to maintain oxygen saturation >94%.
   - **Raise the head of the bed 30°** unless hypotensive or otherwise contraindicated.
   - **For obstructive lung pathology**: Use lower respiratory rates, lower PEEP, and longer I:E ratios to permit time for exhalation; monitor closely for auto-PEEP/air trapping; do not attempt to decrease hypercapnia through increasing Respiratory Rate.
   - **For restrictive lung pathology**: Use higher respiratory rates with greater PEEP and shorter I:E ratios to cater for decreased lung compliance.
   - **For cardiac arrest during CPR:** Use ventilator via an ET tube; it is acceptable to use the ventilator via a supra-glottic airway if functioning correctly and no gastric inflation is observed. Significant care must be taken if using a ventilator via a supra-glottic airway during ambulance movement.
   - **If ETCO2 readings cannot be obtained**, or low minute volumes are observed, consider changing to a Bag Valve Mask.
   - Once ROSC is achieved, adjust ventilator settings appropriate for the patient's underlying lung pathology (normal/obstructive/restrictive lungs).
3. **Ventilation in Pediatrics**
   - Respiratory Rate and Tidal Volume must be adjusted according to pediatric weight and age, using IBW.
   - Other settings remain unchanged.
4. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.
   - Continuously reassess airway status, ventilation, and response to treatment using attached monitors (SpO2, ECG, EtCO2).
   - Adjust settings as needed based on patient condition and lung pathology, ensuring capnography targets (e.g., EtCO2 35-45 mmHg) are met.

Patient Safety Considerations

- Risk of barotrauma or pneumothorax: Use appropriate tidal volumes (6-8 mL/kg IBW for normal/obstructive, 4-6 mL/kg IBW for restrictive) and monitor Pmax (<30-50 cmH2O depending on lung type).
- Hypoxemia or hypercapnia: Ensure proper FiO2 adjustment (>94% saturation) and confirm ventilation with EtCO2 to avoid complications, especially in obstructive lungs where auto-PEEP may occur.
- Aspiration risk: Raise the head of the bed 30° unless contraindicated to prevent aspiration during ventilation.
- Over-ventilation risk: Avoid excessive respiratory rates or tidal volumes, particularly in obstructive lungs, to prevent barotrauma or auto-PEEP.
- Infection risk: Maintain sterile technique during intubation and regular ventilator circuit changes.
- Pediatric safety: Adjust Respiratory Rate and Tidal Volume per weight and age using IBW to prevent over-ventilation or inadequate support.
- Cuff pressure monitoring: When available, use a cuff pressure pneumonitor to maintain ideal cuff pressure to prevent tracheal injury.

Clinical Insights & Additional Information

- Obstructive Lung Pathology
  - Obstructive lung pathology requires lower rates and PEEP to avoid air trapping; do not increase Respiratory Rate to manage hypercapnia.
- Restrictive Lung Pathology
  - Restrictive lung pathology requires higher rates and PEEP for compliance.
- Cardiac Arrest
  - During cardiac arrest, ensure proper ET or supra-glottic airway function; switch to Bag Valve Mask if needed.
- Pediatric Ventilation
  - Pediatric ventilation requires weight- and age-adjusted settings using IBW.
- Ideal Body Weight (IBW) Calculation:
  - Use IBW for tidal volume calculations to ensure lung protection.
  - For males: IBW (kg) = 50 + 0.91 × (height in cm - 152.4).
  - For females: IBW (kg) = 45.5 + 0.91 × (height in cm - 152.4).
- Ventilation Modes:
  - IPPV (Intermittent Positive Pressure Ventilation): Delivers a set tidal volume with each breath, controlled by the ventilator, suitable for patients with consistent respiratory needs.
  - SIMV (Synchronized Intermittent Mandatory Ventilation): Combines mandatory breaths with patient-triggered breaths, allowing partial spontaneous breathing, useful for transitioning off ventilation.

Documentation

- Complete a comprehensive electronic Patient Care Record (ePCR), capturing the following elements to support continuity of care. Including:
  - Identification: Full patient name, age, and gender.
  - Mechanism / Medical Complaint: Presenting condition (e.g., cardiac arrest, respiratory failure), lung pathology.
  - Injuries / Information: Relevant history (e.g., trauma, chronic lung disease).
  - Signs:
    - Full vital signs including pulse, respiratory rate, SpO2, EtCO2.
    - Two sets of vital signs unless critical patient condition precludes.
    - Auscultation findings (e.g., wheezes, crackles).
  - Treatment:
    - Ventilation settings and adjustments.
    - All interventions: Intubation, ventilator use, Bag Valve Mask, and response.
    - Justification if interventions were withheld.
  - Allergies: Documented allergies and drug interactions.
  - Medications: All pre-existing medications and any administered in the field.
  - Background: Relevant medical history, comorbidities, and recent procedures.
  - Other:
    - Communication with Telemedicine (time and instructions).
    - Response to interventions.
    - Consideration of reversible causes and actions taken.

Key Performance Indicators

1. Document initial ventilator settings for all patients where a ventilator is used.

References

1. Ambulance Victoria (2023) *Clinical Practice Guidelines for Ambulance and MICA Paramedics*. Melbourne: Ambulance Victoria, pp. 112–115.

2. Clinical Decision Support Tools for Emergency Care Providers (no date) South Africa: Emergency Care Society of South Africa, p. 18.
3. HMCAS (2023) *Clinical Practice Guidelines*. [Place of publication not specified]: HMC Ambulance Service, p. 167.
4. London Ambulance Service NHS Trust (2022) *Clinical Practice Guidelines*. London: London Ambulance Service NHS Trust, pp. 89–92.
5. NAEMSO (2020) *National Model EMS Clinical Guidelines*. [Online] National Association of Emergency Medical Services Officials. Available at: https://nasemso.org/projects/model-ems-clinical-guidelines/.
6. National Institute for Health and Care Excellence (NICE) (2021) *Emergency and acute medical care in over 16s: service delivery and organisation*. [Online] NICE Guideline NG94. Available at: https://www.nice.org.uk/guidance/ng94.
7. South Western Ambulance Service NHS Foundation Trust (2024) *Clinical Guidelines for Paramedics*. Exeter: South Western Ambulance Service NHS Foundation Trust, pp. 145–148.
8. Tatham, K. C., Patel, K. P. and Sinha, I. P. (2020) 'Mechanical ventilation, clinical practice and clinical decision support systems', *Intensive Care Medicine Experimental*, 8(Suppl 1), p. 44. Available at: https://doi.org/10.1186/s40635-020-00330-2.
9. Tobin, M. J. and Manthous, C. A. (2024) 'Mechanical ventilation', *Respiratory Medicine*, 230, p. 107672. Available at: https://doi.org/10.1016/j.rmed.2024.107672.

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